Setting Up a Custom Inventory Manager Using Firebase and React

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Introduction:

This guide is designed for developers with a basic understanding of JavaScript and React who want to create a Firebase-backed inventory manager. Following these instructions will help you streamline inventory management, suitable for small businesses or personal projects. Each step is broken down to guide you through cloning a GitHub repository, configuring Firebase, and customizing the project.

Required Tools:

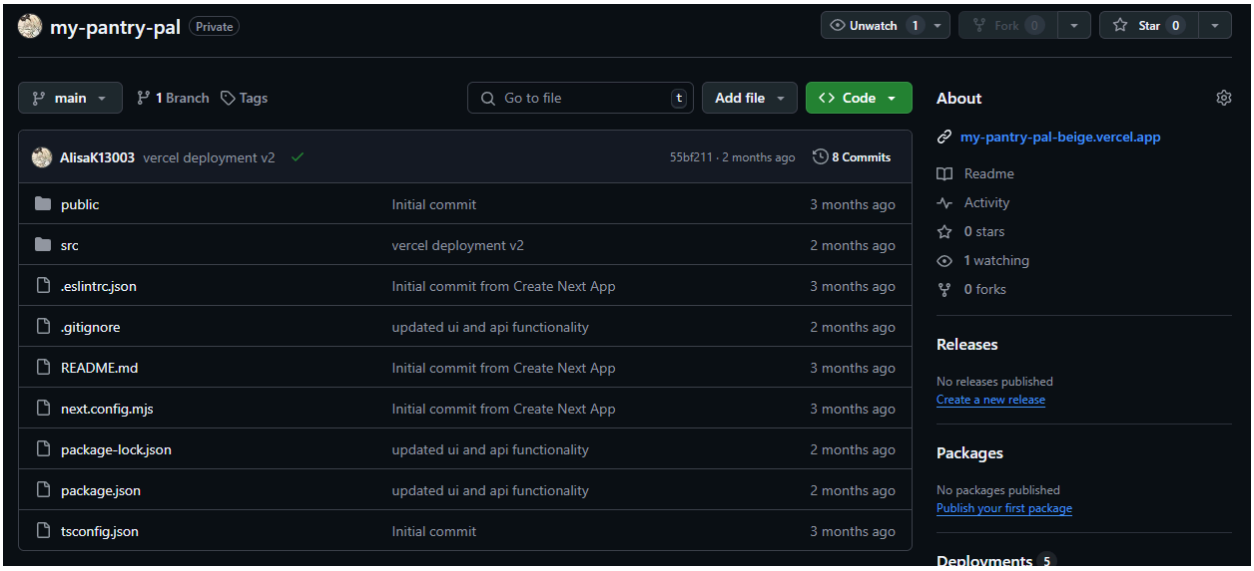
- GitHub account
- Firebase account (sign up at firebase.google.com)
- Node.js (download at nodejs.org)
- Text editor (e.g, Visual Studio Code)
- Basic knowledge of JavaScript and React

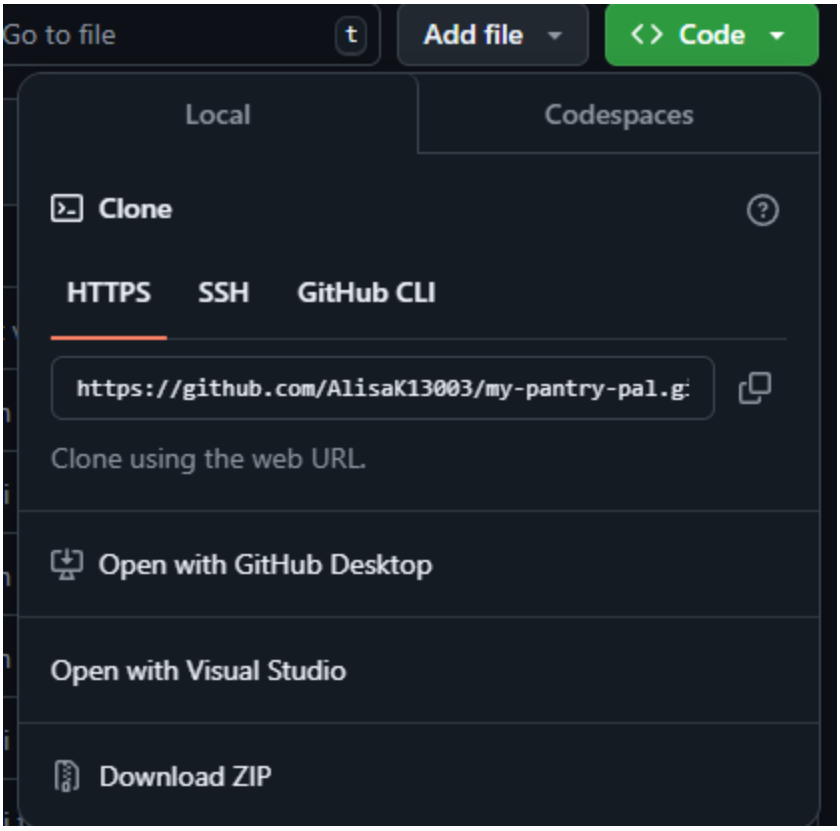
Safety Notes:

- **Firebase Billing**: Ensure you're using the free plan to avoid unexpected charges.
- **API Key Privacy**: Keep your Firebase API keys private, as they contain sensitive information.

"Quick Start" Guide:

1. Clone the [GitHub repository](#).



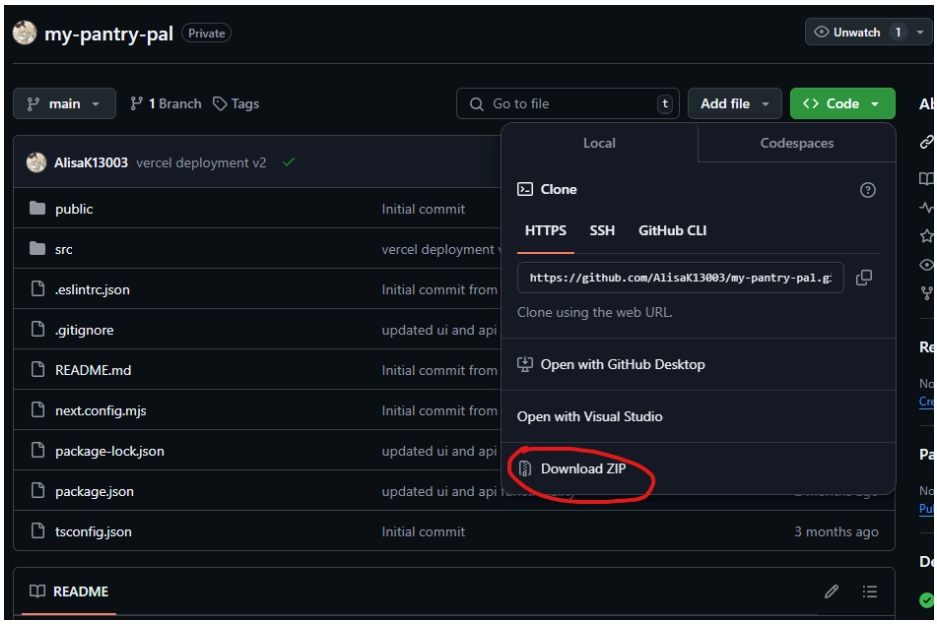


2. Install project dependencies.
3. Set up Firebase following the steps below.

Steps:

Step 1: Clone the Repository (Download the Project Files)

- First, go to the GitHub page for the project: [GitHub Repo Link]
- Click the **"Code"** button and select **"Download ZIP"**. This will download the project files to your computer.



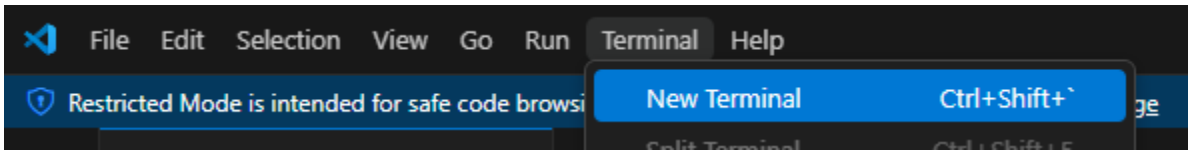
- Unzip the downloaded file to create a folder containing the project files.
    - If you're comfortable with the terminal, you can use:

```
git clone https://github.com/AlisaK13003/my-pantry-pal
```

- Open the project folder in your text editor.

Step 2: Install the Project Dependencies

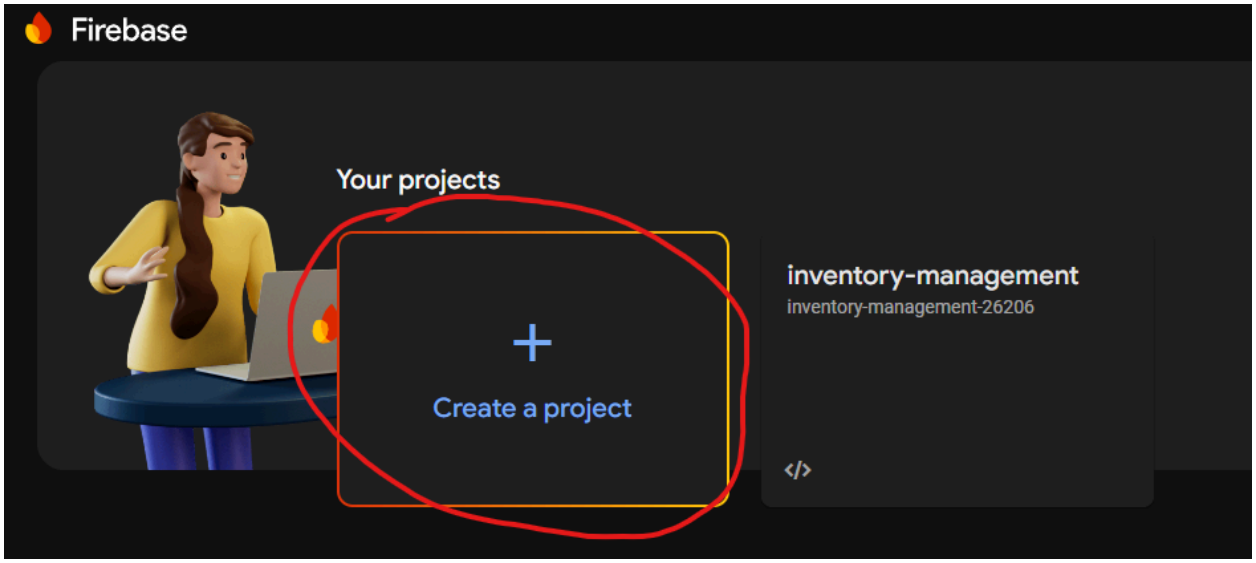
- Open the unzipped folder in your text editor (e.g., Visual Studio Code).
- Open the terminal:
    - In VSCode, click **"Terminal"** in the top menu and select **"New Terminal"**.



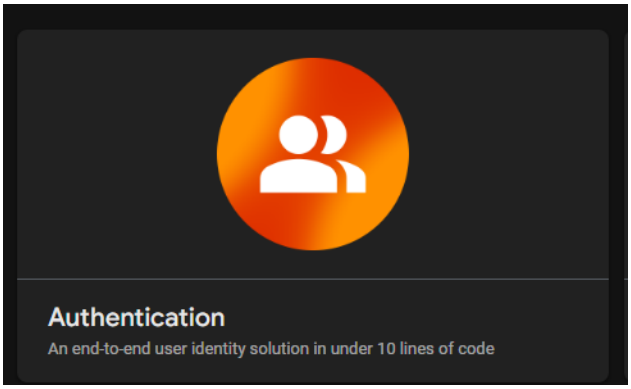
- In the terminal, type the following command to install everything needed for the project:

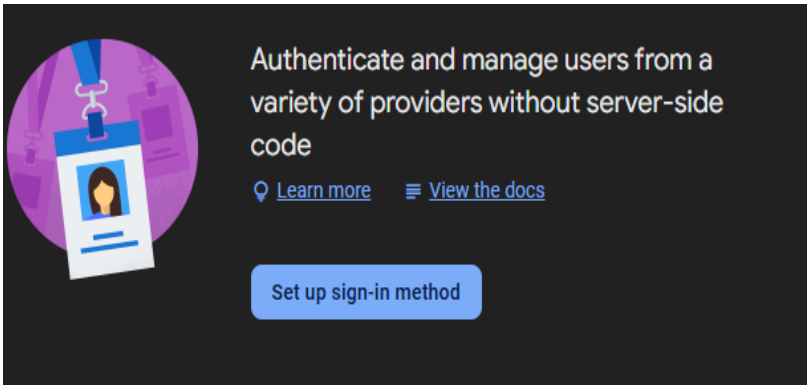
Step 3: Set Up Firebase

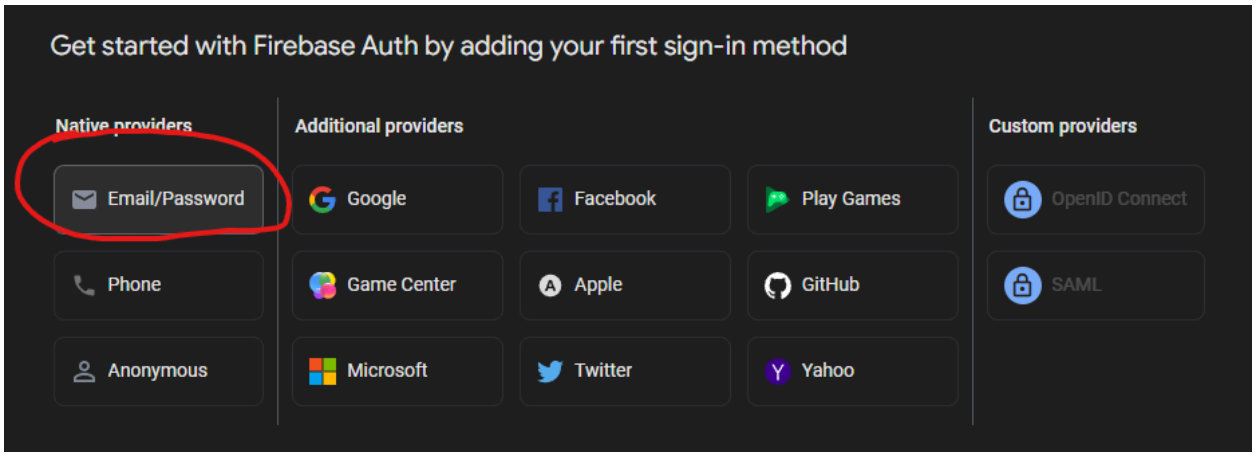
- Go to Firebase Console and create a new project.



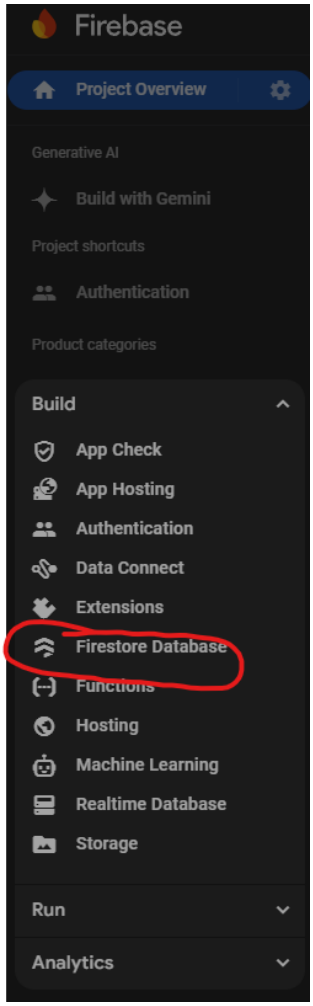
- **Enable Authentication**:
    - In the Firebase dashboard, click on “Authentication” in the left-hand menu.
    - Select “Get Started” and enable “Email/Password” authentication.

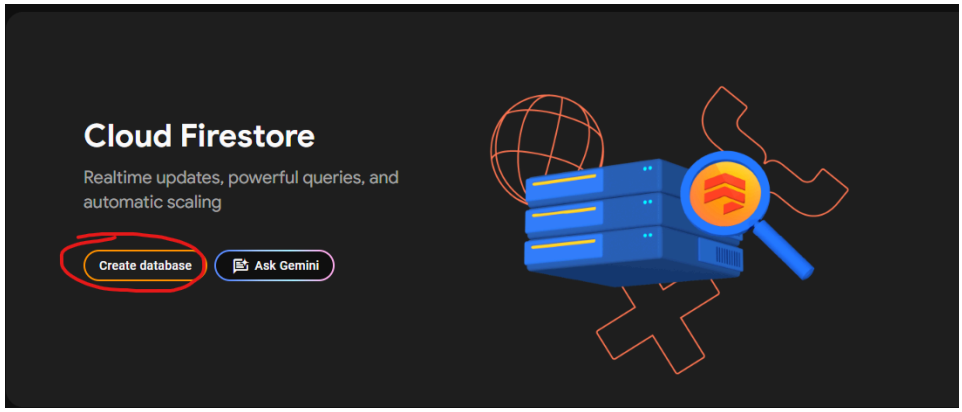


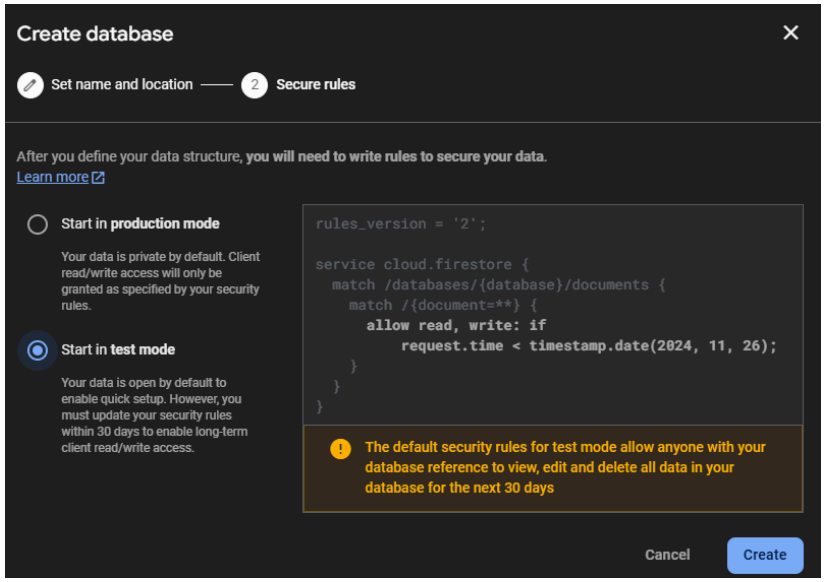




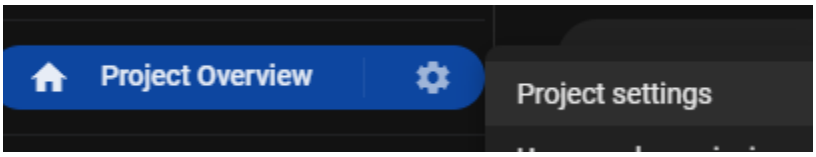
- **Set Up Firestore Database**:
    - In the left-hand menu, click on “Firestore Database.”
    - Click “Create Database” and choose “Start in test mode” to keep things simple.
- After setting up Firebase, go to your project settings in Firebase and find your Firebase configuration information (API key, project ID, etc.).

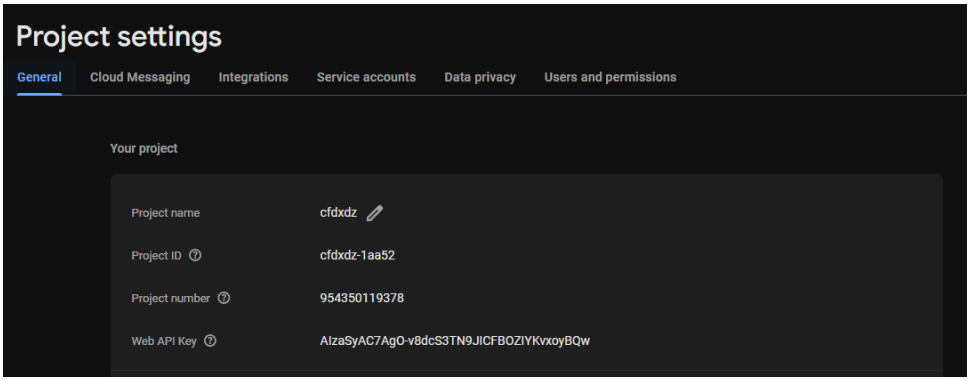






- **Retrieve Firebase Configuration**
  - In Firebase, navigate to your project settings to find your Firebase configuration details (API key, project ID, etc.).





Step 4: Configure Firebase in the Project

- In the project folder you downloaded, create a new file called `.env.local`.
- Open `.env.local` and enter your Firebase configuration, replacing placeholder text with your details:

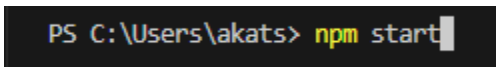
```
REACT_APP_FIREBASE_API_KEY=your_api_key

REACT_APP_FIREBASE_AUTH_DOMAIN=your_auth_domain

REACT_APP_FIREBASE_PROJECT_ID=your_project_id
```

- Save the file.
- Restart your project in the terminal by typing:

```
npm start
```



Step 5: Customize the Application

- Open the **src/components** folder in your project to access main components.
- Modify files to change how inventory items are displayed or add new features by editing the code.

```
import React from 'react';
import Box from '@mui/material/Box';
import Typography from '@mui/material/Typography';
import { useTheme } from '@mui/material/styles';

const Footer = () => {
  const theme = useTheme();

  return (
    <Box sx={{
      display: 'flex',
      justifyContent: 'center', // Centers the inner box
      padding: theme.spacing(2),
      backgroundColor: theme.palette.primary.main, // Ensures the footer background is the primary color
      color: theme.palette.common.white,
      width: '100%', // This ensures it spans the full width of the viewport
    }}>
      <Box sx={{
        display: 'flex',
        justifyContent: 'space-between',
        width: '100%', // Takes the full width of the inner container
        maxWidth: 1200, // Maximum width of the content
        padding: '0 20px', // Padding inside the container
      }}>
        <Typography variant="body2">
          © Copyright My Pantry Pal
        </Typography>
        <Typography variant="body2">
          Designed by Alisa Katasonova
        </Typography>
      </Box>
    </Box>
  );
};

export default Footer;
```

Conclusion

Congratulations! You've successfully set up a custom inventory manager using Firebase and React. Feel free to continue customizing the application and explore Firebase's features to expand functionality.

Troubleshooting Guide:

- **Firebase Error:** Make sure you copied the Firebase configuration details correctly into the `.env.local` file.
- **Database Permission Issues:** If you can't access the Firestore database, check the security rules in the Firebase console. Starting in "test mode" allows public access temporarily.